

Labyrinth Sequencer



Cont mix
seq 1/2

Root /
Scale

Bit sel
seq 1/2

Show
seq 1/2

Select
seq 1/2



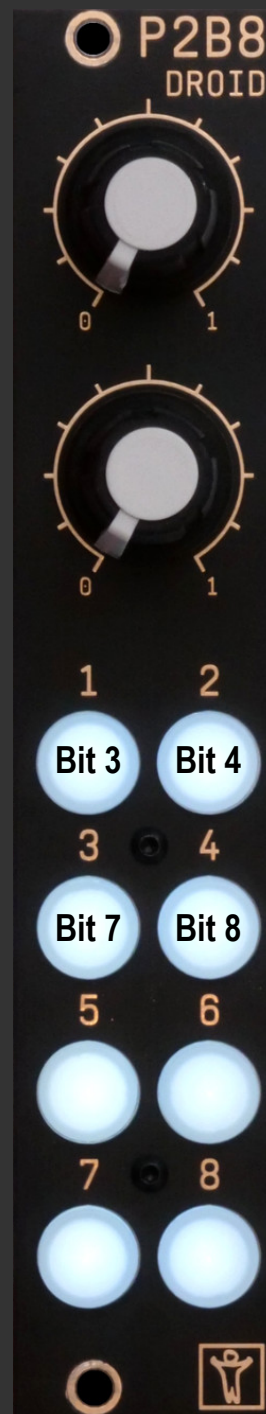
Cont decay
seq 1

Cont decay
seq 2

root /
scale

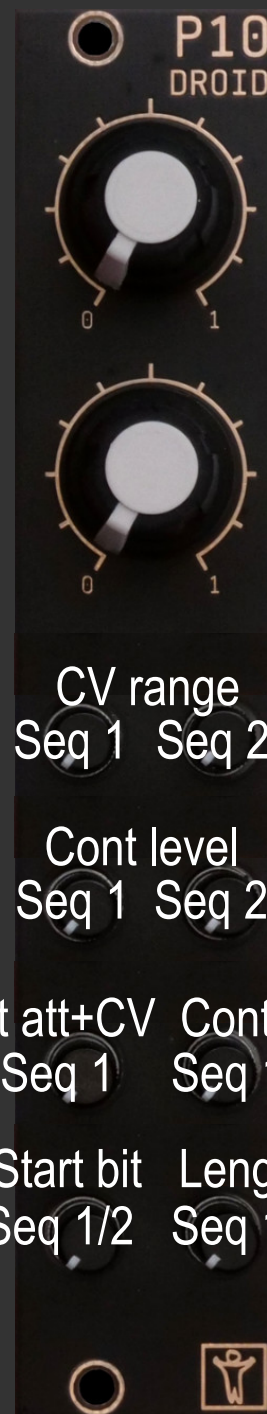
Bit flip
seq 1

Bit flip
seq 2



Corrupt
seq 1

Corrupt
seq 2



CV range
Seq 1 Seq 2

Cont level
Seq 1 Seq 2

Cont att+CV
Seq 1 Cont att
Seq 1+2

Rotate
seq 1/2

Start bit
Seq 1/2 Length
Seq 1/2

Presets

Labyrinth Sequencer

This patch is inspired by the sequencer section of the Moog Labyrinth percussive synthesizer. The main idea is based on two generative sequencers of max 8 bits (steps) that are able to evolve over time in a Turing machine fashion, while also becoming interactively modified by each other. The resulting output stream is characterized by rhythmic and melodic structures that consist of both, repeating and varying elements, making this approach a powerful concept to create lively sequences on the fly without x0x style programming.

Every bit of the sequencers provides a percussion envelope (0-attack, variable decay) and pitch cv output, intentionally being wired to a complex VCO (carrier or modulator), VCF, VCA, wave folder or noise generator to produce a wide spectrum of intriguing percussive sounds. The sequencers can be configured to either randomly change their pitch cv with adjustable probability, or additionally turn individual bits on and off. Interactive modification enables active bits of one sequencer to flip bits of the other sequencer. For more musical control over random creations, individual bits can be resampled, both sequencers can have individual lengths and start bits, and active bits can be rotated. Moreover, new random changes can be set to zero to freeze a current sequence. The pitch cv range can be adjusted manually or by external cv, and the created pitch cv values can be quantized to different scales. The two sequencers can also be chained to a single 16 bit sequence, while still being available twice, with independent clocking, lengths and start bits.

Input jacks:

I1, I5: sequencer 1 clock, sequencer 2 clock
I2, I6: sequencer 1 contour decay, sequencer 2 contour decay
I3, I7: sequencer 1 corrupt, sequencer 2 corrupt
I4, I8: sequencer 1 pitch cv range, sequencer 2 pitch cv range

Suggestions for additional inputs:

- reset (trigger), which is here provided via MIDI
- contour mix (cv),
- sequencer 1 bit flip (trigger), sequencer 2 bit flip (trigger): find example in the patch file for bit flip triggers instead of cv input for decay
- enable control of the contour attack phase or build a more versatile ADSR contour

Output jacks:

O1, O5: sequencer 1 trig, sequencer 2 trig
O2, O6: sequencer 1 contour, sequencer 2 contour
O3, O7: sequencer 1 pitch cv, sequencer 2 pitch cv
O4: sum of sequencer 1 contour and pitch cv, with attenuverter for contour
O8: maximum of sequencer 1 and 2 contours, with attenuverter

Activating sequencer bits (steps):

Each of the two sequencers have max 8 bits, represented by P2B8 buttons. Individual bits can be activated or muted by button short presses.

Bits:	1	2	3	4
	5	6	7	8
Buttons:	B1.1	B1.2	B2.1	B2.2
	B1.3	B1.4	B2.3	B2.4

These buttons are shared by both sequencers. Pressing button B1.5 switches between sequencer 1 and 2. Both sequencers have dedicated clock input jacks at the droid master. Steps with an active bit send a trigger signal (O1, O5), an attack-decay envelope (O2, O6) and a pitch cv value (O3, O7) to associated output jacks.

Envelope mix output:

The envelope depth of the two sequencers can be mixed by pot P1.1, with full intensity of both envelopes at center position. The full intensity, however, is adjustable for each sequencer by the bipolar pots P3.5 and P3.6, respectively, which function as attenuverters, allowing to scale or invert the envelope shape by a maximum factor of +/-0.5. In addition to the individual envelope outputs of sequencers 1 and 2 (O2, O6), a combined envelope output (O8) is provided as the maximum of both envelopes mixed together, and modified by another attenuverter pot P3.8.

Envelope decay:

The output envelopes have a traditional percussion shape with 0 attack and adjustable decay. The decay time can be controlled for sequencer 1 by pot P2.1 or by external cv to input I2, and for sequencer 2 by pot P2.2 or input I6.

Pitch cv range:

The pitch cv values for each sequencer bit (O3, O7) are randomly selected, and further change randomly depending on the probability set by the corruption function (see below). The range of the random pitch cv values is centered around +3V, and can be controlled by pots P3.3 and P3.4 for sequencer 1 and 2, respectively. The maximum range at fully right pot position is from -3V to +3V, providing cv values between 0V and +6V, whereas at fully left pot position, the range is zero, providing a constant cv of +3V (+ an offset from the root note, if quantization is on). A new random pitch cv can be sampled for individual bits by a long press of the sequencer bit buttons. External cv control can be achieved by inputs I4 and I8.

Pitch cv quantisation:

To achieve more tonal sequences, the generated pitch cv can be quantised to particular scales. If button B2.5 is off (dark), pot P1.2 can be used to select a scale by the degree parameter of the minifonion circuit (see droid manual for supported scales). If button B2.5 is on (lit), pot P1.2 sets the root note among 11 semitones. Quantisation is activated or bypassed by the state of button B2.7.

Combined output of sequencer 1 envelope and pitch cv:

Another output is provided as the sum of the envelope and pitch cv of sequencer 1 (O4). Here, attenuverter pot P3.7 only affects the envelope contribution of the signal. Hence at center pot position, the envelope contribution is zero, leaving only the pitch cv signal identical to the original pitch cv output of sequencer 1 (O3).

Sequence corruption:

Random changes of active sequencer bits and their pitch cvs are controlled by pots P3.1 and P3.2 for sequencer 1 and 2, respectively. In the left half of the pots, the probability of changes in pitch cv increases from 0% (fully left) to 50% (center), while the active bits remain the same, thus maintaining the rhythmic structure of the sequences. In the right half of the pots, the probability of pitch cv changes stays at 50% but the probability for changes in active bits increases from 0% (center) to 50% (fully right). At fully right pot position, both the pitch cvs and bit states keep changing with a probability of 50%, whereas at fully left pot position, any changes are omitted, making the sequences strictly repetitive. External control can be achieved by cv input to I3 and I7.

Bit flips:

Another way to let the sequences change dynamically is the bit flip function. Here, every active bit of one sequencer is flipping the state of the bit currently addressed by the other sequencer, enabling coupled sequencer interaction. Button B1.6 activates that sequence 1 is being flipped by sequence 2, and button B1.8 enables sequence 2 being flipped by sequence 1.

Sequence length:

The length of each sequence can be adjusted by pot P3.10. Button B1.7 controls by it's 3 states whether the pot addresses

sequencer 1 (dark),
sequencer 2 (dimmed), or
both sequencers at once (lit),

allowing for synchronized as well as polyrhythmic length settings. Moreover, this parameter can be used to chain both sequences to a single sequence of max 16 bits. By increasing the length beyond 8 bits, the bits of one sequencer will be extended by the bits of the other sequencer, while keeping its individual start bit and contour settings. This enables using two 16 bit sequences that are musically related to each other due to partially shared bit states and pitch cvs.

Sequence start bit:

The start bit of each sequence can be adjusted by pot 3.9, with button B1.7 again selecting the sequencer(s) to which the current settings are applied to.

Sequence rotation:

Each of the two sequences can be rotated. Every press of button B2.6 moves the sequence forward by one bit, with the sequencers being addressed again by Button B1.7.

Presets:

The resulting sequences and corresponding parameter settings can be saved in presets. Button B2.8 enters the preset menu, which turns the bottom two button rows into 7 preset slots (i.e. B1.5-B1.8 and B2.5-B2.8) and the exit button B2.8. Within the preset menu, a long press on a preset slot will save a preset, and a short press will load a preset. The patch file can easily be modified to whether a load signal will also force the sequencers to reset to the saved start bits or just let them continue to run with the new parameters, ignoring set start bits.